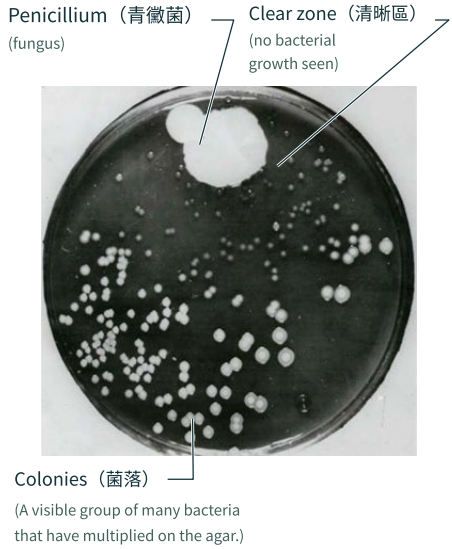
Name: _____ Class: _____ No.: _____

Antibiotic（抗生素） investigation

Record your initial observations. Let your questions guide the study.

The history of antibiotic（抗生素） discovery—an unexpected observation



In 1928, Fleming noticed contamination by Penicillium（青黴菌） on plates growing staphylococci, with reduced bacterial growth nearby. He investigated an antibacterial substance produced by this fungus. Later, Florey, Chain and their team advanced the therapeutic use of penicillin（青黴素）, an antibiotic（抗生素）.

Note: Fleming’s historical work involved staphylococci, not MRSA（耐甲氧西林金黃葡萄球菌）; MRSA（耐甲氧西林金黃葡萄球菌） was first formally reported in 1961.

Image source: https://www.cbsnews.com/news/almanac-the-discovery-of-penicillin（青黴素）/

Research context

A patient’s wound infection has not clearly improved after initial antibiotic（抗生素） treatment. Laboratory testing identifies methicillin-resistant Staphylococcus aureus (MRSA)（耐甲氧西林金黃葡萄球菌）. As a researcher, investigate how samples X, Y and Z affect visible growth of these bacteria in a virtual laboratory.

What is methicillin-resistant Staphylococcus aureus（耐甲氧西林金黃葡萄球菌）?

Resistant（具抗藥性的）: The bacteria resist a particular antibiotic（抗生素）, so it does not effectively kill them or inhibit（抑制） their growth.

Methicillin（甲氧西林）: The name of an antibiotic（抗生素） formerly used to treat some bacterial infections.

Methicillin-resistant Staphylococcus aureus (MRSA)（耐甲氧西林金黃葡萄球菌）: Staphylococcus aureus（金黃葡萄球菌） bacteria that resist methicillin（甲氧西林）. MRSA（耐甲氧西林金黃葡萄球菌） does not resist every antibiotic（抗生素）.

Main research question

Under the stated simulation conditions, how do X, Y and Z affect visible growth of these MRSA（耐甲氧西林金黃葡萄球菌） bacteria?

Your initial observations

Describe what you can see, without explaining the cause yet.

Make a prediction and consider the variables, assumptions and control before starting the experiment.

Complete your prediction

If discs containing antibiotic（抗生素） samples X, Y and Z are placed on an agar plate（瓊脂板）,（X ／ Y ／ Z ／ X and Y ／ X and Z ／ Y and Z ／ X, Y and Z ／ none of the）discs will have a clear zone（清晰區） (a region with no visible bacterial growth) around them.

Among these,（X ／ Y ／ Z ／ Not applicable）will have the largest clear zone（清晰區） around its disc.

Not applicable (no clear zones（清晰區） predicted).

Circle one option in each set of brackets. Preserve your original responses.

Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)

Identify the variables

Choose the suitable factors in each category. Save your thinking now and view reference feedback after submission.

Tick all applicable boxes; more than one option may be selected.

Independent variable（The factor you change）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Dependent variable（The outcome you observe/measure）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Controlled variables（The factors kept the same）

- ☐ Disc size
- ☐ Composition of the agar plate（瓊脂板）
- ☐ Type of bacteria used
- ☐ Preset sample preparation
- ☐ Sample in the disc
- ☐ Visible clear zone（清晰區） and its total diameter
- ☐ Initial bacterial distribution

Your original research plan

Each independent plate has X, Y, Z and the control. One more plate gives one more observation per sample. A higher number does not automatically mean a better plan.

There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)

（1 ／ 2 ／ 3 ／ 4 ／ 5 ／ 6 ／ 7 ／ 8 ／ 9 ／ 10 ／ 11 ／ 12 ／ 13 ／ 14 ／ 15 ／ 16 ／ 17 ／ 18 ／ 19 ／ 20）

Why would you plan it this way?

Preserve the original design in A before distributing B. Do not overwrite it.

What are the assumptions of this investigation?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Select all applicable assumptions; you may choose more than one.

- ☐ Bacteria are initially distributed evenly on each agar plate（瓊脂板）.
- ☐ All plates use the same simulated observation conditions.
- ☐ Equipment does not introduce contamination.
- ☐ If no colonies（菌落） form near a disc, this means all bacteria at that location have died.

The investigation control

If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?

(Hint: The control group provides a baseline for comparison. It helps check whether changes in the experiment are caused by the independent variable rather than other external factors. Therefore, the control group does not receive the treatment being tested.)

(Hint: how does the control differ from the experimental group?)

Draw and label your plates

Use the materials below as a reference. Label X, Y, Z and C (control), considering spacing between discs and from the plate edge. The beaker of alcohol and alcohol lamp are used to sterilise（滅菌／消毒） forceps, reducing contamination when picking up and placing discs.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.



Agar plate（瓊脂板） ×3
(contains nutrients needed for bacterial growth)



Marker ×1



Dropper ×1



Diluted（已稀釋的） bacterial suspension（細菌懸液） ×1



Sterile（無菌的） spreader（塗布器） ×1



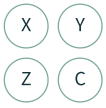
Forceps ×4



Beaker of alcohol ×1



Alcohol lamp ×1



Sample and control disc set ×3
(C = control)



Incubator（培養箱） ×1



Ruler ×1

Setup labels / written design

Add labels to your drawing, or give a complete written description of your plates as an alternative to drawing.

Preserve the original design in A before distributing B. Do not overwrite it.